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Ubuntu Server (Redirector)

Objective

This document describes the installation, configuration, and validation of the Ubuntu Server virtual machine that serves as the redirector within the Red Team lab.

The redirector forms part of the operator's infrastructure and is intended to relay operational traffic between the operator workstation and compromised systems during later phases of the engagement.

Role Within the Lab

The Ubuntu Server functions as the lab's redirector. Its primary role is to provide an intermediary system through which operational traffic can later be relayed, reducing direct exposure of the operator workstation to the target environment.

During the initial build, the redirector is prepared as a standard Ubuntu Server installation. Additional services and configurations required for traffic relaying, secure remote management, and C2 operations will be introduced during subsequent phases of the lab.

Design Considerations

The redirector is intentionally deployed as a dedicated virtual machine rather than combining its responsibilities with the operator workstation.

Separating these roles provides a clearer understanding of how intermediary infrastructure can be used during offensive operations and allows each machine to serve a single, well-defined purpose.

The redirector should remain lightweight, easily reproducible, and disposable. If the infrastructure is no longer required or becomes unusable during experimentation, it should be possible to rebuild it without affecting the remainder of the lab.

Installation

This section documents the deployment of Ubuntu Server as a virtual machine.

Download Ubuntu server .iso file from it web

Download VirtualBox

Open VirtualBox

Go to Home

Click on New

Enter VM Name

Leave Folder as default or set your preferred location

Select the .iso downloaded

Uncheck “Proceed with Unattended Installation”

Go to “Specify virtual hardware”:

Set “Base Memory” to 4096 or leave default

Set CPU to two or leave default

Leave “Use EFI” unchecked

Set Disk size to 50GB or leave default

Click “Finish”

Start the server

Step 2:

How does Ubuntu physically enter the Cloud Network, and how does pfSense then give Ubuntu an IP address?

Set your Ubuntu Redirector Network to Internals; then choose the network for the redirector network.

(In my case Cloud’s Network, which is for OPT1).

Power On Your Ubuntu

Select your language

Select your keyboard type

Installation Type: Ubuntu Server

Network Configuration: I set mine manually

```
Network configuration [ Help ]

Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient
access for updates.

NAME      TYPE  NOTES
[ enp0s3  eth  -           ▶ ]
static    172.16.10.10/24
08:00:27:2a:d1:4f / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter)

[ Create bond ▶ ]
```

—— Edit enp0s3 IPv4 configuration ——

IPv4 Method: [Manual ▼]

Subnet: 172.16.10.0/24

Address: 172.16.10.10

Gateway: 172.16.10.1

Name servers: 172.16.10.1
IP addresses, comma separated

Search domains:
Domains, comma separated

[Save]
[Cancel]

Proxy Configuration: leave blank

Mirror Configuration: I leave mine blank

Storage Configuration: Leave default

Profile Configuration:

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: IntelliJSage

Your servers name: ubuntu-svr
The name it uses when it talks to other computers.

Pick a username: IntelliJSage

Choose a password: *****

Confirm your password: *****

Password: @ubuntu-svr1

Upgrade to Ubuntu Pro: Skip for now (unless you have token for it)

SSH Configuration: Install SSH

Error Encountered

Connectivity issue:

I tried to reach a server (1.1.1.1) outside my network but failed. And I tried reaching the pfSense (172.16.10.1) still fails.

```
File Machine View Input Devices Help
intellisage@ubuntu-svr:~$ ip route
default via 172.16.10.1 dev enp0s3 proto static
172.16.10.0/24 dev enp0s3 proto kernel scope link src 172.16.10.10
intellisage@ubuntu-svr:~$ ping -c 4 172.16.10.1
PING 172.16.10.1 (172.16.10.1) 56(84) bytes of data.

--- 172.16.10.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3655ms

intellisage@ubuntu-svr:~$ _
```

Initial Configuration

Document the basic operating system configuration performed immediately after installation.

Examples include:

- Updating installed packages.
 - Configuring the hostname.
 - Setting the system time.
 - Installing VirtualBox Guest Additions (if used).
 - Creating a clean snapshot after successful installation.
-

Validation

Verify that the operating system has been installed successfully and is functioning as expected.

Typical validation checks include:

- Successful system boot.
- User login.
- Package update completed successfully.
- Network adapter detected.
- Internet connectivity (if required for package installation).
- Snapshot created.

Any issues encountered during installation should be resolved before proceeding to later configuration stages.

Observations

Record any notable findings encountered during installation or initial configuration.

Examples include hardware compatibility issues, VirtualBox configuration adjustments, or installation decisions made during the build.

Questions

What is the meaning of adapter?

Why are you configuring two adapters?

How is Kali able to connect to Ubuntu server when they are both not on the same network?

Lessons Learned

Summarise the key concepts learned while deploying the Ubuntu Server virtual machine and explain how those concepts support the overall objectives of the Red Team lab.

Changes Timeline

Date	Change	Reason
30/07/2026	Initial Ubuntu Server deployment documented	Establish baseline redirector configuration